TRAUMA · NEET-SS / FET / INI-SS

Ankle fractures

1. Applied anatomy

- The ankle mortise: tibial plafond, medial malleolus, lateral malleolus (fibula, the main lateral restraint). The **talus is wider anteriorly** (stable in dorsiflexion).
- **Ramsey and Hamilton: 1 mm of lateral talar shift reduces tibiotalar contact area by about 42%** (the basis for anatomical reduction).
- **Medial ligament (deltoid): superficial** (crosses ankle and subtalar; tibionavicular, tibiospring, tibiocalcaneal) and **deep** (deep anterior and posterior tibiotalar; **the primary restraint to lateral talar shift**).
- **Lateral ligaments: ATFL** (weakest, commonest sprain; tight in plantarflexion), **CFL** (crosses ankle and subtalar joints), PTFL (strongest).
- **Syndesmosis: AITFL**, **PITFL** (strongest component, with its deep part the inferior transverse ligament), interosseous ligament, interosseous membrane.
- The posterior malleolus is attached to the PITFL; a posterior malleolar fracture can destabilise the syndesmosis.

2. Classifications

Lauge-Hansen (mechanism: position of the foot + direction of force)

Type	Frequency	Sequence of injury	Fibular fracture
Supination-external rotation (SER)	**Commonest (about 40-75%)**	I AITFL, II **short oblique/spiral fibula at the level of the syndesmosis** (anteroinferior to posterosuperior), III PITFL or posterior malleolus, IV medial malleolus fracture or deltoid rupture	Weber B
Supination-adduction (SAD)	About 10-20%	I transverse fibular avulsion below the joint (or lateral ligament rupture), II **vertical medial malleolus fracture** with **medial plafond impaction** (fix and elevate)	Weber A
Pronation-external rotation (PER)	About 7-20%	I medial malleolus/deltoid, II AITFL, III **high spiral fibula fracture above the syndesmosis**, IV PITFL/posterior malleolus	Weber C (includes **Maisonneuve**)
Pronation-abduction (PAB)	About 5-20%	I medial malleolus/deltoid, II syndesmosis/posterior malleolus, III **comminuted/transverse fibula fracture at or above the syndesmosis** (lateral comminution)	Weber B/C

- Remember: the **first word is the foot position**, the second the force; the **first injured structure is on the tension side** (supination = lateral first; pronation = medial first).

Danis-Weber (level of the fibular fracture relative to the syndesmosis)

- **A:** below the syndesmosis (infra-syndesmotic); syndesmosis intact.
- **B:** at the level of the syndesmosis (trans-syndesmotic); syndesmosis may be injured.
- **C:** above the syndesmosis; syndesmosis torn (needs assessment and usually fixation).

Special patterns

- **Maisonneuve fracture: proximal fibula fracture + syndesmotic and interosseous membrane disruption + medial injury** (PER). Always palpate the whole fibula in an "isolated" medial malleolar fracture or deltoid injury. Treat with **syndesmotic fixation** (fibula itself usually not fixed).
- **Bosworth fracture-dislocation:** fibula **trapped behind the posterolateral tibia (incisura)**; irreducible by closed means: **emergency open reduction**.
- **Tillaux-Chaput** (AITFL avulsion from the tibia) and **Wagstaffe-Le Fort** (AITFL avulsion from the fibula) fragments.
- **Posterior malleolus** fractures: **Haraguchi classification** (CT): I posterolateral oblique (commonest), II medial extension, III small shell (rim). **Bartoníček-Rammelt** classification relates to the incisura.

3. Assessment

- **Ottawa ankle rules** (radiograph if pain in the malleolar zone plus tenderness at the posterior edge/tip of either malleolus, or inability to bear weight four steps immediately and in the department; foot series if tenderness at the base of the fifth metatarsal or navicular).
- **Radiographs:** AP, lateral, **mortise view** (15-20 degrees internal rotation).

  - **Medial clear space: over 4-5 mm** (or greater than the superior clear space) = **deltoid injury / lateral talar shift**.
  - **Tibiofibular clear space** (AP, 1 cm above the plafond): **over 5-6 mm = syndesmotic injury** (most reliable radiographic measure).
  - **Tibiofibular overlap:** under 6 mm on AP or under 1 mm on mortise = syndesmotic injury.
  - **Talocrural angle** (about 83 degrees ±4): assesses fibular length; **dime sign** (continuous curve between the fibular tip and lateral talus) shows fibular length.
- **Stress views: gravity or manual external rotation stress** (medial clear space over 5 mm) or **weight-bearing radiographs** decide whether an isolated Weber B (SER) fracture is stable: **a weight-bearing mortise with a normal medial clear space indicates stability** even if the gravity stress view widens.
- **CT** for posterior malleolar fractures, pilon-type impaction, and syndesmosis assessment; **MRI** for occult injuries.

4. Management

Stable injuries (non-operative)

- **Weber A**, **isolated undisplaced Weber B with a stable mortise** (normal medial clear space on weight-bearing film), isolated undisplaced medial malleolar fractures.
- Treatment: **walking boot or cast with early weight bearing as tolerated** for 6 weeks. The **AIM trial** (elderly, unstable ankle fractures, close-contact casting vs ORIF) showed **equivalent function at 6 months** with skilled casting; **CROSSBAT** (Kortekangas): undisplaced/minimally displaced Weber B can be treated non-operatively (casting for 3 weeks equals 6 weeks).

Unstable injuries (operative)

- **Bimalleolar**, **trimalleolar**, unstable SER IV equivalents (fibula + deltoid with talar shift), syndesmotic injuries, **open fractures**,

irreducible dislocations, **talar shift**.

- **Emergency: reduce a dislocated ankle immediately** (skin and neurovascular compromise), splint, elevate; ORIF when swelling allows (within about 24 hours or after 7-10 days).
- **Fibula:** restore **length and rotation**: lag screw with **one-third tubular neutralisation plate (lateral)** or **posterolateral antiglide plate** (Weber B; biomechanically stronger, less hardware prominence, risk to peroneal tendons distally). Intramedullary fibular nail in poor soft tissue/elderly.
- **Medial malleolus: two 4.0 mm partially threaded cancellous screws** perpendicular to the fracture, or tension-band wiring for small fragments; **vertical (SAD) fractures need horizontal screws or an antiglide plate**, plus elevation of medial plafond impaction. The **deltoid ligament usually heals without repair** once the fibula is reduced (repair if the medial clear space does not reduce: interposed deltoid or tibialis posterior tendon; some advocate repair in high-demand athletes).
- **Posterior malleolus:** fix when **over about 25-33% of the articular surface** (traditional rule), when there is **posterior talar subluxation**, an **intercalary/impacted fragment**, or when it destabilises the syndesmosis. Current evidence favours **direct fixation (posterolateral approach) of larger fragments** because it restores syndesmotic stability and incisura congruity better than a syndesmotic screw.
- **Syndesmosis:**
  - **Intraoperative tests: Cotton (hook) test** (lateral pull on the fibula with a bone hook: over about 2 mm widening), external rotation stress.
  - **Fixation: screw(s) (3.5 or 4.5 mm, through 3 or 4 cortices, 2-5 cm above the joint, parallel to the plafond, angled about 25-30 degrees anteriorly from posterior)** or **suture-button (dynamic) fixation**. **Suture-buttons** give fewer malreductions and less need

for removal (better functional scores in some trials); screws may be removed after about 3 months (debated).
    - **Accurate reduction is critical**: **direct visualisation of the incisura** or intraoperative CT, because **malreduction (up to about 30-50% on CT)** is the main cause of poor outcome. Clamp along the transmalleolar axis to avoid over-compression or malrotation.
- **Post-operative:** early range of motion; weight bearing in a boot from about 2-6 weeks (early weight bearing is safe in most fixed fractures).

Special groups

- **Diabetic patients (especially with neuropathy):** high complication rates (infection, Charcot, loss of fixation). **"Super-construct" fixation** (longer plates, multiple syndesmotic screws, transarticular pins), prolonged non-weight bearing (about double the usual time).
- **Elderly osteoporotic:** locking plates, fibular nails, **tibiotalocalcaneal (hindfoot) nail** for very poor bone/soft tissue; skilled close-contact casting (AIM).
- **Open fractures:** debridement, antibiotics, stabilisation (external fixation or internal fixation depending on contamination), soft-tissue cover.

5. Ankle sprains and related injuries

- **Lateral ankle sprain:** ATFL then CFL. **Anterior drawer** (ATFL), **talar tilt** (CFL). **Functional treatment** (brace, early motion, proprioceptive rehabilitation) for all grades acutely.
- **Chronic lateral ankle instability** after failed rehabilitation: **anatomical repair (Broström-Gould: ATFL/CFL repair reinforced with the inferior extensor retinaculum)**; anatomical reconstruction with tendon graft for poor tissue, failed repair,

generalised laxity or high BMI; non-anatomical (Chrisman-Snook, Watson-Jones, Evans) rarely now (stiffness of the subtalar joint).

- **High ankle sprain (syndesmosis): squeeze test**, external rotation (Kleiger) test; longer recovery; stabilise if unstable on stress/arthroscopy.
- **Missed injuries around a "sprain": lateral process of the talus fracture** (snowboarder's), **anterior process of the calcaneus fracture**, **base of fifth metatarsal fracture**, **osteochondral lesion of the talus**, **peroneal tendon subluxation/tear**, **Maisonneuve fracture**.
- **Osteochondral lesions of the talus: anterolateral** (shallow, wafer-shaped, traumatic) vs **posteromedial** (deeper, cup-shaped, often atraumatic). **Berndt-Harty** staging; MRI/CT; **arthroscopic debridement and microfracture/bone-marrow stimulation for lesions under about 1.5 cm^2 (or under 15 mm diameter)**; osteochondral autograft (OATS) or other cartilage procedures for larger lesions.

6. Complications of ankle fractures

- **Post-traumatic arthritis** (malreduction of fibular length/rotation, syndesmotic malreduction, articular damage).
- **Wound complications and infection** (smokers, diabetes, early surgery through swollen skin).
- **Syndesmotic malreduction**, **broken syndesmotic screws** (usually harmless), loss of reduction.
- **Superficial peroneal nerve** injury (lateral incision, about 7 cm above the tip of the fibula it pierces the fascia), sural nerve (posterolateral approach), saphenous nerve (medial).
- Complex regional pain syndrome, stiffness, DVT.

Exam pearl: **SER is the commonest Lauge-Hansen type (fibula at the syndesmosis, anteroinferior to posterosuperior). PER IV gives a high fibular fracture; Maisonneuve = proximal fibula + syndesmotic disruption. 1 mm talar shift = 42% loss of contact area.**

Exam pearl: **Medial clear space over 4-5 mm = deltoid injury; tibiofibular clear space over 5-6 mm = syndesmotic injury.** Weight-bearing radiographs decide stability of an isolated Weber B. **Syndesmotic malreduction is the main cause of poor outcome; visualise the incisura.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.